

Correction of Exercises in LaTeX

Subject code: Corr_exo

TODO: author name(s)

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Exercise 3.1: Existence of Implicit Schemes

Let $\Psi : \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be Lipschitz in all variables. Denote by L a Lipschitz constant of Ψ with respect to its third variable. For fixed (t, x, h) , define

$$\mathcal{T}(y) = x + h\Psi(t, x, y).$$

If $hL < 1$, then $\mathcal{T}$ is a contraction because

$$|\mathcal{T}(y_1) - \mathcal{T}(y_2)| \leq hL|y_1 - y_2|.$$

By the Banach fixed point theorem, the equation

$$y = x + h\Psi(t, x, y)$$

has a unique solution. A numerical method is Picard iteration:

$$y^{(0)} = x, \quad y^{(k+1)} = x + h\Psi(t, x, y^{(k)}).$$

For $hL < 1$, this sequence converges geometrically to the unique solution $s(t, x, h)$.

Now suppose $y = s(t, x, h)$ and $y = x + h\phi(t, x)$. Then

$$\phi(t, x) = \frac{s(t, x, h) - x}{h} = \Psi(t, x, s(t, x, h)).$$

This is well-defined because $s(t, x, h)$ is unique. To prove that s is Lipschitz, let $s = s(t, x, h)$ and $\tilde{s} = s(\tilde{t}, \tilde{x}, h)$. Then

$$|s - \tilde{s}| \leq |x - \tilde{x}| + hL(|t - \tilde{t}| + |x - \tilde{x}| + |s - \tilde{s}|).$$

Thus, when $hL < 1$,

$$|s - \tilde{s}| \leq \frac{(1 + hL)|x - \tilde{x}| + hL|t - \tilde{t}|}{1 - hL}.$$

So s is Lipschitz. Since Ψ is Lipschitz and $\phi(t, x) = \Psi(t, x, s(t, x, h))$, the function ϕ is also Lipschitz for sufficiently small h .

Exercise 3.2: Euler-Lagrange Multipliers

Let

$$L(\lambda, x, y) = f(x, y) + \lambda g(x, y).$$

Since $y_0 = Y(x_0)$, we have

$$g(x_0, y_0) = 0.$$

Also $\partial_y g(x_0, y_0) \neq 0$, so the equation

$$\partial_y L(\lambda_0, x_0, y_0) = \partial_y f(x_0, y_0) + \lambda_0 \partial_y g(x_0, y_0) = 0$$

has the unique solution

$$\lambda_0 = -\frac{\partial_y f(x_0, y_0)}{\partial_y g(x_0, y_0)}.$$

Therefore

$$\partial_y L(\lambda_0, x_0, y_0) = 0, \quad \partial_\lambda L(\lambda_0, x_0, y_0) = g(x_0, y_0) = 0.$$

Differentiate the constraint $g(x, Y(x)) = 0$ at x_0 :

$$\partial_x g(x_0, y_0) + \partial_y g(x_0, y_0) Y'(x_0) = 0.$$

Also

$$F'(x_0) = \partial_x f(x_0, y_0) + \partial_y f(x_0, y_0) Y'(x_0).$$

Using $\partial_y f(x_0, y_0) = -\lambda_0 \partial_y g(x_0, y_0)$ gives

$$\begin{aligned} F'(x_0) &= \partial_x f(x_0, y_0) - \lambda_0 \partial_y g(x_0, y_0) Y'(x_0) \\ &= \partial_x f(x_0, y_0) + \lambda_0 \partial_x g(x_0, y_0) \\ &= \partial_x L(\lambda_0, x_0, y_0). \end{aligned}$$

For $f(x, y) = x + y$ and $g(x, y) = x^2 + y^2 - 1$, the assumption $\partial_y g(x_0, y_0) \neq 0$ means $y_0 \neq 0$. We obtain

$$\lambda_0 = -\frac{1}{2y_0}.$$

Thus

$$F'(x_0) = \partial_x L(\lambda_0, x_0, y_0) = 1 + \lambda_0(2x_0) = 1 - \frac{x_0}{y_0}.$$

This computes the derivative without explicitly solving for the local branch $Y(x)$.

For the vector case, let

$$f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad g : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n,$$

and define

$$L(\lambda, x, y) = f(x, y) + \lambda^\top g(x, y), \quad \lambda \in \mathbb{R}^n.$$

Assume $D_y g(x_0, y_0)$ is invertible. The multiplier is determined by

$$\nabla_y f(x_0, y_0) + D_y g(x_0, y_0)^\top \lambda_0 = 0,$$

so

$$\lambda_0 = -D_y g(x_0, y_0)^{-\top} \nabla_y f(x_0, y_0).$$

Differentiating $g(x, Y(x)) = 0$ gives

$$D_x g(x_0, y_0) + D_y g(x_0, y_0) DY(x_0) = 0.$$

As in the scalar case, the terms containing $DY(x_0)$ cancel and

$$\nabla_x F(x_0) = \nabla_x f(x_0, y_0) + D_x g(x_0, y_0)^\top \lambda_0 = \nabla_x L(\lambda_0, x_0, y_0).$$

Exercise 3.3: Multi-Step SDE Scheme

Consider

$$Y_{n+1} = Y_n + \frac{3}{2}a_n h - \frac{1}{2}a_{n-1}h + b_n\sqrt{h}\xi_n,$$

where ξ_n is independent of the filtration $\mathcal{A}_{t_n}$, has mean m , and has variance σ^2 .

For weak consistency, the conditional mean of the increment must match the drift to first order. Since

$$\mathbb{E}[Y_{n+1} - Y_n \mid \mathcal{A}_{t_n}] = \frac{3}{2}a_n h - \frac{1}{2}a_{n-1}h + b_n\sqrt{h}m,$$

the term $b_n\sqrt{h}m$ would be too large compared with the drift unless

$$m = 0.$$

With $m = 0$, the Adams–Bashforth drift satisfies

$$\frac{3}{2}a_n - \frac{1}{2}a_{n-1} = a_n + o(1),$$

because a is smooth and Y_{n-1} is close to Y_n as $h \rightarrow 0$.

For the second moment,

$$\mathbb{E}[(Y_{n+1} - Y_n)^2 \mid \mathcal{A}_{t_n}] = b_n^2 h \mathbb{E}[\xi_n^2] + o(h).$$

Since $m = 0$, $\mathbb{E}[\xi_n^2] = \sigma^2$. Matching the diffusion coefficient requires

$$\sigma^2 = 1.$$

Therefore the scheme is weakly consistent when

$$\boxed{m = 0, \quad \sigma^2 = 1.}$$

The scheme is not strongly consistent in general with merely discrete variables ξ_n . Strong consistency is pathwise and requires the noise increment to match the Brownian increment in mean square. A standardized discrete variable may have the correct first two moments, but it is not the same object as $\Delta W_n/\sqrt{h}$. The mean-square difference between $b_n\sqrt{h}\xi_n$ and $b_n\Delta W_n$ does not vanish at the local order needed for strong consistency unless ξ_n is chosen to be the actual normalized Brownian increment. Thus the given weak scheme is generally not strongly consistent.